

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2281

Cranberry 138 kV
March 31, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2281. The Interconnection Customer proposes a 105.3 MW battery energy storage facility interconnecting at the Cranberry 138 kV substation in the Pennsylvania area. The request was received on January 24, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 678 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,244,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yucca Substation to Fraley 138.0 kV L	Sugar Hill to Allegheny Energy Units 3 4 & 5 345.0 kV L	123.5	383	5.1
Houseville to Fraley 138.0 kV	Sugar Hill to Allegheny Energy Units 3 4 & 5 345.0 kV	123.3	391	5.1
Va Hospital to Houseville 138.0 kV Line	Sugar Hill to Allegheny Energy Units 3 4 & 5 345.0 kV	123.2	395	5.1
Sugar Hill to Va Ospital 138.0 kV	Sugar Hill to Allegheny Energy Units 3 4 & 5 Substation 345.0 kV	123.1	403	5.1
Youngwood 500/345kV Transformer	Southeast Vicksburg to Allegheny Energy Units 3 4 & 5 500.0 kV	105.9	873	5.5
Allegheny Energy Units 3 4 & 5 345/138kV XFMR	Allegheny Energy Units 3 4 & 5 345-1 kV XFMR	100.1	438	5.9
Craberry 500 kV/345 kV Transformer Bank	Stewart Mountain Dam to Ritter Park 345.0 kV	100.0	916	26.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yucca Substation to Fraley 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,019,000	16% (\$471,000)
Fraley to Houseville 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,016,000	16% (\$783,000)
Va Hospital to Houseville Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,383,000	16% (\$372,000)
Sugar Hill to Va Hospital 138.0 kV	Rebuild line with higher-capacity conductor	\$6,916,000	16% (\$1,080,000)
Youngwood 500/345kV Transformer	Replace transformer with higher-rated unit	\$36,094,000	5% (\$1,943,000)
Allegheny Energy Units 3 4 & 5 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$15,918,000	39% (\$6,193,000)
Craberry 500 kV/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$31,908,000	58% (\$18,402,000)
	Total allocated to project		\$29,244,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.